

Yu-Wei Fan

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Research Interests

Formal verification, automated reasoning, and hardware security.

Education

Ph.D. in Electrical and Computer Engineering

Princeton University, USA

2024–Present

Advisor: Prof. Sharad Malik

M.S. in Electronics Engineering

National Taiwan University (NTU), Taiwan

2024

Advisor: Prof. Jie-Hong Roland Jiang

B.S. in Electrical Engineering

National Taiwan University (NTU), Taiwan

2022

Research Experience

Research Assistant

Princeton University, Sharad Malik's Group

2024–Present

Advisor: Prof. Sharad Malik

- Researching automated refinement of taint propagation logic for hardware IFT.
- Researching hardware model checking algorithms for information-flow properties.

Research Assistant

National Taiwan University, Applied Logic and Computation Lab

2020–2024

Advisor: Prof. Jie-Hong Roland Jiang

- Researching stochastic Boolean satisfiability solving and its extensions.
- Researching reverse engineering of gate-level arithmetic circuits.

Adjunct Research Assistant

Institute of Information Science, Academia Sinica

2022–2023

Advisor: Dr. Yu-Fang Chen

- Applied software testing techniques, including concolic testing, to DNN models.

Publications

* indicates equal contribution

Conference Papers

- [C1] **Yu-Wei Fan**^{*}, Yuheng Yang^{*}, Christine Guo, SooHyuk Cho, Thomas Bourgeat, Mengjia Yan, and Sharad Malik. “CEGAR-T: Automated Taint Logic Optimization for Hardware Information Flow Tracking”. Under review.
- [C2] Tingzhen Dong^{*}, Kunpeng Wang^{*}, Yuheng Yang, **Yu-Wei Fan**, Qinhan Tan, Thomas Bourgeat, Sharad Malik, and Mengjia Yan. “Interplay of Efficient Model Checking and Secure Processor Design: A Case Study on Secure Speculation”. In: *IEEE Symposium on Security and Privacy (SP)*. 2026.
- [C3] Kuo-Wei Ho, Jie-Hong Roland Jiang, Alan Mishchenko, Sean A. Weaver, and **Yu-Wei Fan**. “Hierarchical Boundary Recovery: Overcoming Synthesis Obscurity via SAT Sweeping”. In: *The Chips To Systems Conference (DAC)*. 2026.
- [C4] Qinhan Tan, Akash Gaonkar, **Yu-Wei Fan**, Sharad Malik, and Aarti Gupta. “SecIC3: Customizing IC3 for Hardware Security Verification”. In: *Design, Automation and Test in Europe Conference (DATE)*. **Best Paper Nomination**. 2026.
- [C5] **Yu-Wei Fan**, and Jie-Hong Roland Jiang. “Unifying Decision and Function Queries in Stochastic Boolean Satisfiability”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2024.
- [C6] Long-Hin Fung, Che Cheng, **Yu-Wei Fan**, Tony Tan, and Jie-Hong Roland Jiang. “2-DQBF Solving and Certification via Property-Directed Reachability Analysis”. In: *Formal Methods in Computer-Aided Design (FMCAD)*. **Best Paper Award**. 2024.
- [C7] **Yu-Wei Fan**, and Jie-Hong Roland Jiang. “SharpSSAT: A Witness-Generating Stochastic Boolean Satisfiability Solver”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2023.
- [C8] Kuo-Wei Ho, Shao-Ting Chung, Tian-Fu Chen, **Yu-Wei Fan**, Che Cheng, Cheng-Han Liu, and Jie-Hong Roland Jiang. “WolFEx: Word-Level Function Extraction and Simplification from Gate-Level Arithmetic Circuits”. In: *International Conference on Computer-Aided Design (ICCAD)*. 2023.

Journal Articles

- [J1] Long-Hin Fung, Che Cheng, **Yu-Wei Fan**, Tony Tan, and Jie-Hong Roland Jiang. “2-DQBF Solving and Certification via Property-Directed Reachability Analysis”. Under review.

Awards and Honors

2026 Princeton ECE Pramod Subramanyan *17 Early Career Graduate Award
2024 Institute of Information and Computing Machinery Master Thesis Award, Honorable Mention
2024 IEEE Taipei Section Master Thesis Award
2024 Taiwanese Association of Artificial Intelligence Master Thesis Award, Honorable Mention
2024 NTU Graduate Institute of Electronics Engineering Master Thesis Award

Teaching Experience

Teaching Assistant, Contemporary Logic Design, Princeton University	Fall 2025
Teaching Assistant, Introduction to Electronic Design Automation, NTU	Spring 2023, 2024
Teaching Assistant, Switching Circuits and Logic Design, NTU	Fall 2022